Start and Stop Delay Constraints in EMG Command Filtering

A Reproducible Replay Benchmark

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Abstract

Background: Filtering of the output from a classifier for movements allows suppression of unwanted activations, but this can add delays in activating and deactivating. In such a scenario, a controller sending out fewer commands can still perform poorly when responsiveness and persistent activation are considered.

Objective: The objective was to assess whether independently chosen activation and release persistence resulted in decreased exposures to the Annotated-Rest commands compared to symmetric persistence while meeting pre-defined criteria for initiation, termination, and active coverage periods.

Methods: The present offline replay benchmark involves the use of the Ninapro DB5 and DB2 datasets. DB5 has two development subjects and eight reserved evaluation subjects, whereas DB2 has eight development subjects and 32 reserved evaluation subjects. For each subject, the decoder was trained using repetitions 1, 3, and 4, while reserved evaluation involved repetitions 2 and 5. Training-only standardization and shrinkage linear discriminant analysis was carried out via causal 150 ms frames with hop sizes of 25 ms. Seven different controller types were involved, totaling 107 setups. The requirement for development feasibility is that each subject achieves at least 90% stability-correct onsets within 250 ms, 90% stability rest confirmations within 150 ms, and 80% nonzero active interior. Fallback configurations were employed for infeasible setups. Differences between paired subjects were assessed using the bootstrap confidence interval method.

Results: The development configuration was unable to satisfy the criteria for all participants in the study. Specifically, DB5's use of an asymmetric fallback led to a rise in mean rest command exposure from 7.217% to 8.295%, which constitutes a difference of +1.077 percentage points (95% confidence interval, 0.764 to 1.397). All 14 configurations using DB5 resulted in none out of eight participants fulfilling the criteria jointly. Meanwhile, in DB2, an increase of +4.165 percentage points (3.463 to 4.904) came about via transporting asymmetries; an independent re-selection process produced +2.572 percentage points (2.186 to 2.963). No joint successes were attained by any of the 19 DB2 configurations among all 32 participants.

Conclusions: The separation of activation and release persistence did not provide an improvement meeting the joint requirements in this benchmark. These results highlight the importance of interpreting rest-command suppression along with confirmed response, halting, and correct-command output. These results relate to the specific offline pipelines examined here and make no claims regarding clinical performance.

Keywords: electromyography; myoelectric control; biomedical engineering; command filtering; response delay; reproducibility

Introduction

The movement classifier outputs a class label. However, a prosthetic control system also needs to determine whether it should initiate action based on that label and when it should terminate action.

These questions are somewhat interrelated, yet they represent different issues. While enforcing agreement might resolve temporary problems with classifier error, it could also prolong inappropriate behavior.

The accuracy-versus-delay trade-off for myoelectric control has been well documented. Smith et al. quantified the effects of increasing analysis window size versus increasing controller delay [1]. Farrell explained the role of windowing and majority voting in causing delay [2]. These studies support measuring response timing rather than merely estimating it from window lengths; they do not establish a universal delay budget across devices/users.

Postprocessing methods, too, have rich histories. Class transition analyses have differentiated transient errors made at movement transitions from sustained error rates in steady-state operation [3]. Decision change informed rejection (DCIR) has leveraged temporal features of the classifier output to modulate rejection thresholds [4]. Onset detection and vote locking have addressed command stability [5,6]. Most recently, confidence gating and temporal voting have both been applied in the same low-cost prosthetic arm pipeline [7]. Thus, temporal rejection and unreliable movement decisions alike are well-understood problems at this point.

Here, we ask a more specific question. Is it possible, with independently specified persistence parameters, to minimize undesired command durations while meeting explicit criteria for initiation, termination, and ongoing control? It is necessary first to determine whether such a design is even feasible, and then to compare its potential exposure times against existing solutions. An entirely quiescent controller may yield minimal rest-command exposure time, although the result would provide negligible control benefit. Similarly, a high percentage of nonzero-command coverage achieved through continual erroneous outputs serves little practical purpose.

Our seven controller designs share a decoder, a fixed set of annotation criteria, and only use development set data during selection trials. The main analysis uses DB5, followed by DB2 native rate replays re-trained on individual calibration data. It is meant to serve as a reproducible examination of the constraints and limitations imposed thereby. Neither a novel controller design nor clinical utility claims are being advanced herein.

Methods

Study design and analytical records

Our study uses publicly available EMG signals for analysis. The initial experimental protocol, dated to October 15th, 2026, at 16:30:42 UTC, and its associated hashed manifest, dated to October 15th, 2026, at 16:32:52 UTC, preceded feature extraction, decoder training, selection of controllers, evaluation of performance, etc. The specific DB5 controller settings were locked at 17:24:15 UTC before generating predictions for reserved subjects. Our second DB2 controller selection was locked at 17:55:57 UTC prior to the generation of DB2 predictions. These dates are internally tracked timestamps for our records and do not represent a preregistered public commitment. Our codebase was released publicly based upon the analyses outlined in reference [12].

For purposes of selecting controllers, we divided our sample into development participants and reserved participants. Decoders were personalized: each reserved subject’s decoder used designated training repetitions, after which the decoder was scored on evaluation repetitions. Predictions for non-calibrated individuals are not evaluated herein. Some training repetitions were recorded at a time subsequent to evaluation repetition 2. Thus, our experiment addresses within-recording replay subsequent to individual calibration but not predictive deployment.

Under the prespecified interpretation rules, neither family of controllers could claim superiority under constraints unless both families met the joint requirements in development phases. Subsequent comparisons maintained this stipulation. No changes were made to the protocol, models, thresholds, data splitting procedures, or final results during manuscript composition.

Datasets and source checks

DB5 consists of recordings from ten subjects with intact limbs, wearing two Myo armbands with 16 EMG channels sampled at 200 Hz [8, 10]. Exercise B was used canonically, consisting of 17 active movement labels alongside a rest label. Subjects with IDs 1 and 2 comprised the development group; those with IDs 3 to 10 comprised the evaluation group. The integrity of the original DB5 archive files was verified via comparison with the reported file size, MD5 hash, archive CRC value, and SHA-256 hash. Original bytes were maintained. Despite the filename referring to E2, the exercise scalar remained as 1. Subject alias 11 from S2 was also retained and verified explicitly.

DB2 includes recordings from 40 subjects with intact limbs, possessing 12 EMG channels at the documented native sampling frequency of 2000 Hz [9, 11]. Canonically, exercise B was extracted from the unmodified E1 set of MATLAB members. Participants were designated by IDs 1 to 8 as development subjects; those between 9 and 40 were the evaluation subjects. All data underwent checks for participant/exercise metadata, channel counts, finite-valued signals, labels, repetitions, and raw-range ownership prior to fitting. Archive verifications included verifying the member length, the CRC value, full extraction, and computing SHA-256 hashes. Provider checksums for the complete archives could not be verified with this download range method.

Both datasets utilized the identical 17 active labels with rest in this task. Neither dataset experienced exclusions based on the ID assignment scheme. The dataset IDs cannot ensure independence of biological participants between datasets. Therefore, DB2 is considered replication across acquisition environments instead of independent biological participants. Table 1 summarizes the experimental setup.

Table 1 Dataset roles and evaluation design

Dataset	Development IDs	Evaluation IDs	Channels and rate	Final events
DB5	1–2	3–10	16 at 200 Hz	272
DB2	1–8	9–40	12 at 2000 Hz	1088

Each participant trained on repetitions 1, 3, and 4. Development scoring used 2, 5, and 6; reserved evaluation used 2 and 5, leaving 6 unused. Each final participant contributed 34 eligible events. Dataset IDs do not establish biological independence across datasets.

Repetition ownership and causal features

For each subject, repetitions 1, 3, and 4 comprised decoder training data. Repetitions 2, 5, and 6 comprised development evaluation, while repetitions 2 and 5 were reserved for evaluation purposes. For reserved evaluation participants, the remaining repetition 6 was held-out and not included in the current investigation. Each bout of movement activity was bounded by an adjacent rest period, wherein the interval midpoint marked the boundary of two adjacent activity bouts. Such half-open intervals non-overlappingly divided up the source recording into blocks. Only contiguous half-open intervals performing identical roles were merged. Feature extraction windows and control-state updating were prevented from traversing real block gaps.

Our feature window was sized at 150 ms and progressed in increments of 25 ms. Given an endpoint e, the corresponding source window was [e−L,e), while the associated label referred to the annotation

value at e−1. The decision time of each source point was e/f_s. Endpoint constraints adhered to a global hop grid requiring full inclusion inside a block. Our setup resulted in feature windows containing 30 samples hopped by 5 for DB5, and windows of 300 samples hopped by 50 samples for DB2. Further filtering and/or resampling was not performed in the primary pipeline. The released signals themselves underwent provider preprocessing including signal alignment and interpolation in the case of DB5 [8]. Causal herein pertains to the feature extraction and control state updating relative to released sample points rather than end-to-end causal acquisition and preprocessing performed by the provider.

Our feature vector included per-channel values for mean absolute value (MAV), root mean square (RMS), and mean waveform length, yielding a total of 48 features in DB5 and 36 in DB2. Given channel samples x_1…x_L, the MAV referred to the average value of |x_i|, RMS referred to sqrt(mean(x_i^2)), and the mean waveform length to sum(|x_(i+1)-x_i|)/(L−1). Inputs were converted to floating point before arithmetic computation. Our input features were arranged on a per-channel basis.

We trained a StandardScaler using the feature windows of only each participant’s training set, employing population variance normalization alongside unit scaler normalization for constant features. Our classifier model was linear discriminant analysis using the lsqr solver algorithm with shrinkage regularization and uniform class prior distribution across all 18 classes. Model fitting and calibration did not employ any evaluation-role windows. Probability outputs from our classifier supplied the policy confidence value; no probability calibration scheme was adopted.

Controller families and selection

Each family adopted equivalent decoder predictions and features. Specifically, we had raw decisions; causal majority voting; symmetric persistence; asymmetric persistence; confidence rejection; decision-change informed rejection (DCIR); and onset locking, the latter of which acted on mean absolute value (MAV) computed via channel data (hence MAV onset locking). Their respective numbers of candidate configurations were 1, 6, 6, 36, 10, 28, and 20, for a total of 107 configurations. Exact values of grids used are detailed in the protocol.

For symmetric persistence, the activation count was equal to the release and change counts. In asymmetric persistence, separate counts were made for activation and release, while the change operation utilized the activation count. Sets of counts included {1, 2, 3, 5, 7, 9}; each time a candidate changed, its confirmation count restarted at one. For majority voting, decision windows were set to {1, 3, 5, 7, 9, 13}. If the current output was tied for the maximum count, it was kept; otherwise, the most recent tied decision was used.

Confidence rejection returned the rest command unless the highest posterior probability exceeded the threshold value by a predetermined strict criterion. DCIR involved setting the threshold to qmin+ (qmax-qmin) exp(-t/τ), where t denoted the elapsed time since the previous change of raw decision state. These definitions, along with the resetting process, remained constant for purposes of comparative analysis. MAV onset locking determined whether the mean absolute value was higher than a threshold equal to the training-rest mean plus a development-selected multiple of the training-rest standard deviation. At or below the threshold, rest and reset occurred; above, initial outputted predictions were sent, after which the selected initial vote count was used to lock the recent plurality state, maintaining this until a new lower amplitude reset criterion was met.

These were all typical implementations of common-decoder methodology applicable to the benchmark, although they cannot be viewed as exact duplicates of all aspects chosen in related hardware implementations, training protocols, or preprocessing strategies of the originals [4]. Any

configuration was viable if all three joint constraints could be fulfilled simultaneously by every developer-participant combination. Configurations sought to minimize mean exposure to rest command; with ties resolved by mean false-activation episode rate, then mean capped onset delay, and finally canonical ordering of parameters. In cases where no configuration proved viable, the frozen fallback configuration sought to minimize the maximum fraction deviation from constraints; followed by average; preserving minimization order above. Such an approach produced a configuration described as infeasible and descriptive. No such configuration would subsequently be considered viable based solely on performance rankings relative to other unsuccessful candidates.

Additionally, five frozen reference benchmarks included: raw output; 9-decision majority voting; confidence rejection at posterior threshold value of 0.97; DCIR qmin=0.4,qmax=0.989,τ=480ms; and six-initial-vote 3σ-MAV onset locking. Also considered were asymmetric persistence sub-families involving only the activation or release functions. The total resulted in 14 distinct DB5 approaches, several of which shared output streams identically.

Held commands and outcome definitions

Each output command was held constant until the next decision or the end of a block. For output, resting was provided before the first decision within a block, and all controller history was cleared at block edges. For each method, the initial 250ms of each block was omitted in computing durations, although causal state updates proceeded normally. Duration scores were computed on the native sample time scale, not on counts of classified frames.

Eligible events had at least 1000ms of activity annotation, preceded by at least 200ms of rest, followed by at least 500ms of rest within the same block, with the onset occurring after the scoring warmup period. Final inventory sizes were exactly 34 events per participant: 272 DB5 events and 1088 DB2 events. All events passed through analysis unchanged regardless of method.

An event had successful onset if its output matched the annotated correct movement for a minimum 100ms span confirmed no later than 250ms post onset. Successful offset occurred if the output remained in the rest state for a minimum 100ms span confirmed no later than 150ms past the annotated offset time. These windows amount to five endpoints when hopped by 25ms increments. Confirmation is applied to the last confirming endpoint and not moved retroactively to the start of the stable span. Offset compliance makes use of all eligible events; a method not exhibiting activity prior to offset does not need to exhibit a transition to resting, as compliance is satisfied if it remains resting afterward. This is marked and the actual values for first-zero and release delays saved based upon previous activity status. The score should be considered alongside these conditional diagnostics and the level of active coverage achieved.

Active coverage refers to the ratio of time spent executing nonzero commands within eligible active periods starting at least 500ms after onset and ending no later than annotated offset. It accounts for both correctly-identified and incorrectly-identified active states. Fractions were given separately for each category (correct, wrong-active, zero-command). Joint success requires at least 90% onset success, at least 90% offset compliance, and at least 80% active coverage by a participant.

The main exposure outcome was the proportion of scored annotated rest periods having nonzero held commands. Therefore, the exposure Ei refers to the proportion of nonzero-command samples within annotated rest periods for participant i. It includes periods of activity that persist after offset. This metric cannot be confused with a false activation rate at the frame-level nor with measured physical activity.

Statistical analysis and sensitivity checks

The main comparison involved the equal-participant mean of 100 times the asymmetric-minus-symmetric exposure difference, expressed in percentage points. It was calculated using 10,000 bootstrap draws with seed 20261015, where the lower bound is the 2.5 percentile and upper bound 97.5 percentile. The DB5 also maintained the frozen two-sided exact sign-flip calculation using all 256 sign permutations. Null interpretation for the DB5 requires the exchangeability of signs; thus, randomized treatment proof cannot be provided here.

Additionally, five secondary comparisons compared the selected asymmetric approach with the baseline raw, majority voting, confidence rejection, DCIR, and MAV locking approaches. The Holm correction was performed on the five comparisons. Owing to feasibility constraints for developing all approaches, all primary and secondary numerical comparisons should be considered descriptive. Low p-values could not confirm the constrained hypothesis. Resample units were the individual participants rather than overlapping windows.

All 23 DB5 sensitivity analyses included feature windows of 100, 200, and 250 ms; stability windows of 50 and 150 ms; shifts of annotations by −100, −50, +50, and +100 ms; stimuli annotated; a constant balanced logistic regression classifier; and the 12 combinations of onset budgets {150, 250, 350, 500} ms and offset budgets {100, 150, 250} ms. Alternative annotations retained the existing raw partitions and checked eligibility. Budget and window conditions followed development choices specified previously. Logistic regression retained the LDA-selected controller settings, with C=1, class_weight="balanced", solver="lbfgs", and max_iter=2000. None of the parameter values were determined based on reserved results. Our sensitivity analysis covers the primary combination (250/150 ms) of budget parameters as well.

Finally, we analyzed a fixed OTVL-inspired adaptation approach post hoc following disclosure of the primary result. It used the same causal 5 Hz envelope; lag-validated peaks; calibration during training only; 150 ms inhibition period; and 8/10 vote locking. Again, it reused the shared classifier as before and was purely exploratory. Different sensing, conditioning, training, and task designs make it impossible to treat this as replicating the entire OTVL system [6].

DB2 replication and computational verification

In DB2, native channels and timing were retained. For the main replication, the seven DB5-selected parameter configurations were transported while fitting each DB2 participant’s decoder on that participant’s training repetitions. In the supplementary replication, the same 107 configurations were reselected using the eight development participants in DB2 before final prediction. Nineteen methods involved combinations of seven transported parameters, seven reselected parameter sets, and five fixed reference configurations. Both replications made use of the identical 32 final participants. In DB2 bootstraps, 10,000 paired draws from the participant pool were performed, without adopting any novel p-value computation method since the pre-computed exact DB5 values only applied to the eight participants.

Verification steps included validation checks on synthetic units, source and execution hashes, native-time reconstruction of held output predictions, event confirmation checks, and numerical verification of results against saved computations. After releasing the runner code for public use, the same procedure was performed again using the publicly released code base to refit and re-run all primary DB5 and final DB2 methods now aware of the outcome. These runs performed reproducibility checks without constituting new, unseen experiments. Sequential sampling profiling assessed the time required for feature extraction, scaling, posterior inference, and policy update computations performed

on the execution host machine. Sensing, communication, and mechanical operations were not assessed.

Results

Development feasibility and selected settings

None of the configurations explored satisfied all three requirements of both development participants for either family. This finding was true for all seven candidate families considered within the primary analysis window as well as the additional windows analyzed a priori. For the selected symmetric fallback strategy, a single decision made, making it analogous to simple outputs. The asymmetric fallback involved one activation decision and two release decisions. Majority voting involved a single decision, while confidence rejection had zero thresholds applied. Thus, the raw, selected majority, selected symmetric, and selected confidence streams were identical in output; repeating the same output does not provide independent confirmation of a result.

The selected DCIR parameters were qmin = 0.2, qmax = 0.8, and tau = 100 ms. Three sigma MAV locking was performed with one initial vote. Activation-only was identical to symmetric, whereas release-only was equivalent to the full asymmetric fallback. No family was found to be budget feasible.

Results of Primary DB5 Evaluation

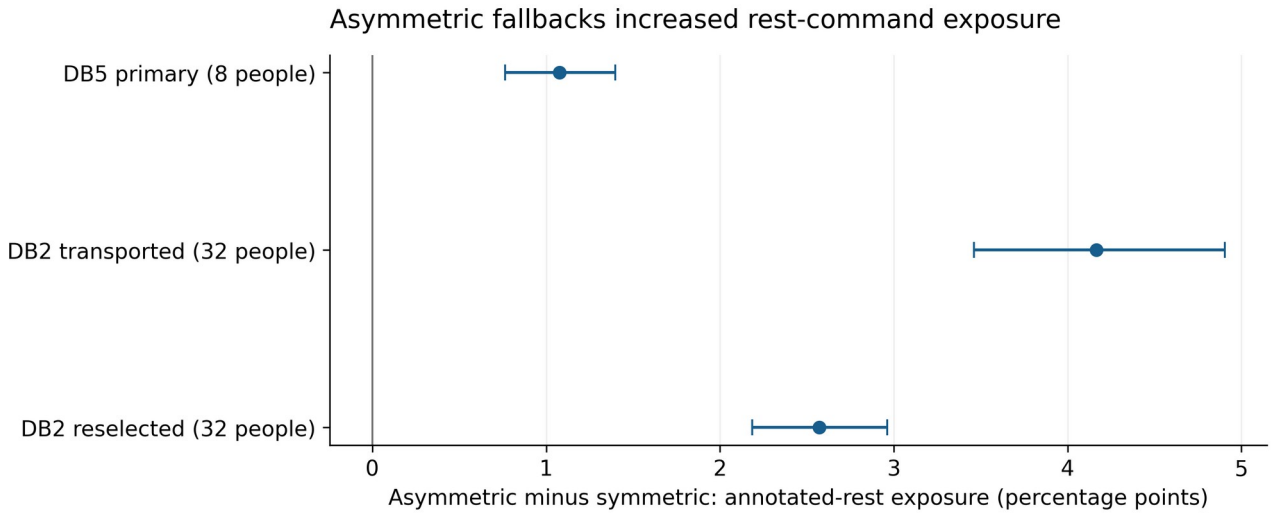
Mean command exposure during annotated rest was 8.295% in asymmetric persistence compared to 7.217% in symmetry. The difference (+1.077 percentage points, 95% bootstrap CI 0.764 to 1.397 percentage points) was positive in all eight participants evaluated. The descriptive exact sign flip probability value was 0.0078125. See Figure 1 for this estimate compared to two previous DB2 contrasts.

Onset success rate in stability correction was 6.99% and 6.25% using the asymmetric vs. symmetric approaches. Offset compliance rates were 33.82% and 37.87%, respectively. Non-zero interior coverages were 94.39% and 93.30%. Despite meeting the mean criteria > 80%, none of the seven families resulted in one participant satisfying joint requirements. All 14 named DB5 configurations yielded a zero success in eight participants.

Table 2 Selected DB5 fallback outcomes

Method	Rest exposure %	Onset success %	Offset compliance %	Active coverage %	Correct interior %
Raw	7.22	6.25	37.87	93.30	57.76
Majority vote	7.22	6.25	37.87	93.30	57.76
Symmetric	7.22	6.25	37.87	93.30	57.76
Asymmetric	8.29	6.99	33.82	94.39	58.10
Confidence rejection	7.22	6.25	37.87	93.30	57.76
DCIR	2.62	1.10	75.00	64.22	49.61
MAV locking	1.90	1.10	66.91	78.70	18.22

Equal-participant means were calculated across eight individuals. Out of the total seven candidate designs considered, none could meet feasibility criteria in development, resulting in 0/8 joint successes when finalized. Pre-defined criteria include 90% success rate of initiating an active state, 90% compliance with termination requirements, and 80% active coverage.



The rest exposure differences are illustrated in Fig. 1. These contrasts are calculated as mean (asymmetric – symmetric) rest exposure for each participant, with 95% confidence intervals determined using the percentile bootstrap. Positive numbers denote a larger proportion of nonzero-command time in asymmetric compared to symmetric conditions. Descriptive statistics only, since the analysis failed feasibility criteria. Both analyses of DB2 use identical sets of n=32 participants.

On comparison against the selected majority and fallbacks under the selected confidence level, the contrast of the exposure asymmetry persisted as +1.077 points on account of symmetry equivalence. Relative to the selected decision-change informed rejection (DCIR), the contrast grew to +5.679 points (3.342 to 8.274); and relative to the selected MAV-locking scheme, it grew to +6.399 points (3.569 to 9.322). Every secondary exact p-value equaled 0.0078125 (corrected by Holm method: 0.0390625). These are descriptive contrasts involving infeasible representatives.

Low exposure did not necessarily mean usable retained activity. Fixing the confidence level threshold to 0.97 led to rest exposure of 0.106%. Interior coverage with non-zero values was 23.25%, and no onset satisfied the primary deadline. Selected MAV locking provided coverage of 78.70%, yet the percentage of correct command for interior was only 18.22%. Incorrectly commanded interiors accounted for 60.47%. Decomposition of results is illustrated in Fig. 2. Correct interior fractions for selected asymmetry and symmetry approaches were 58.10% and 57.76%, respectively.

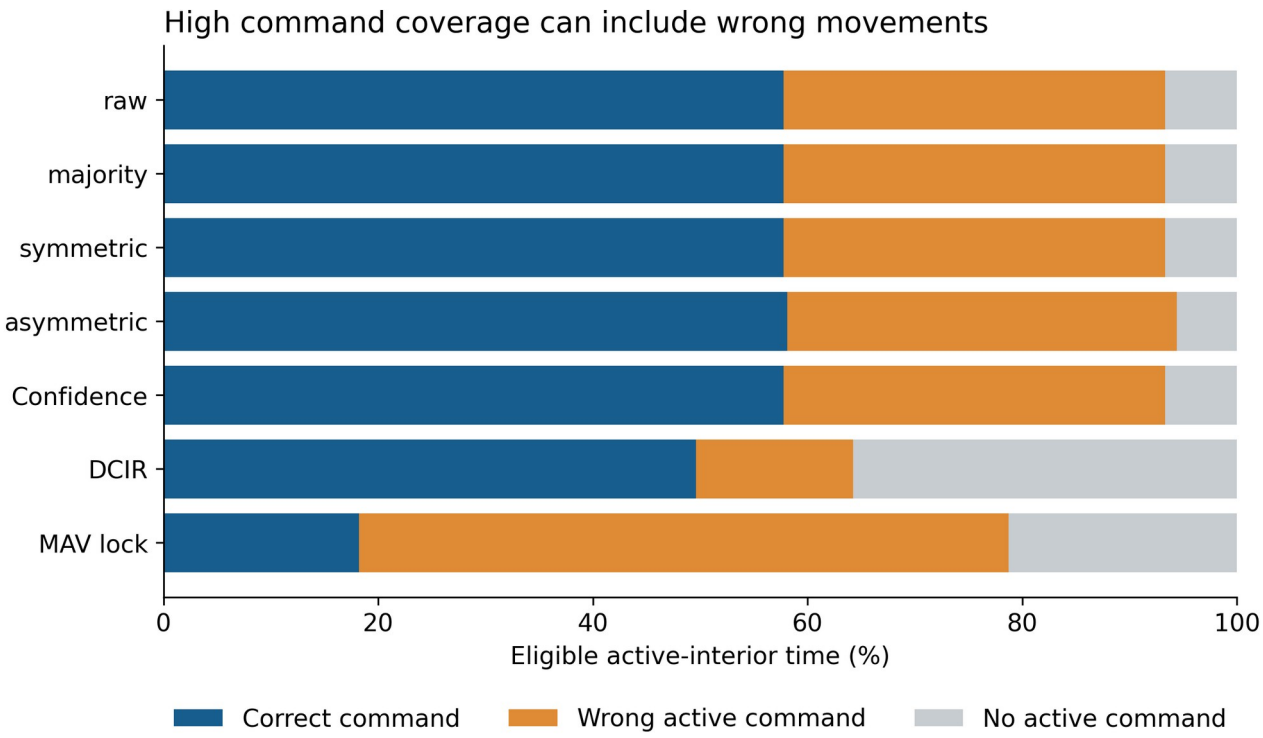


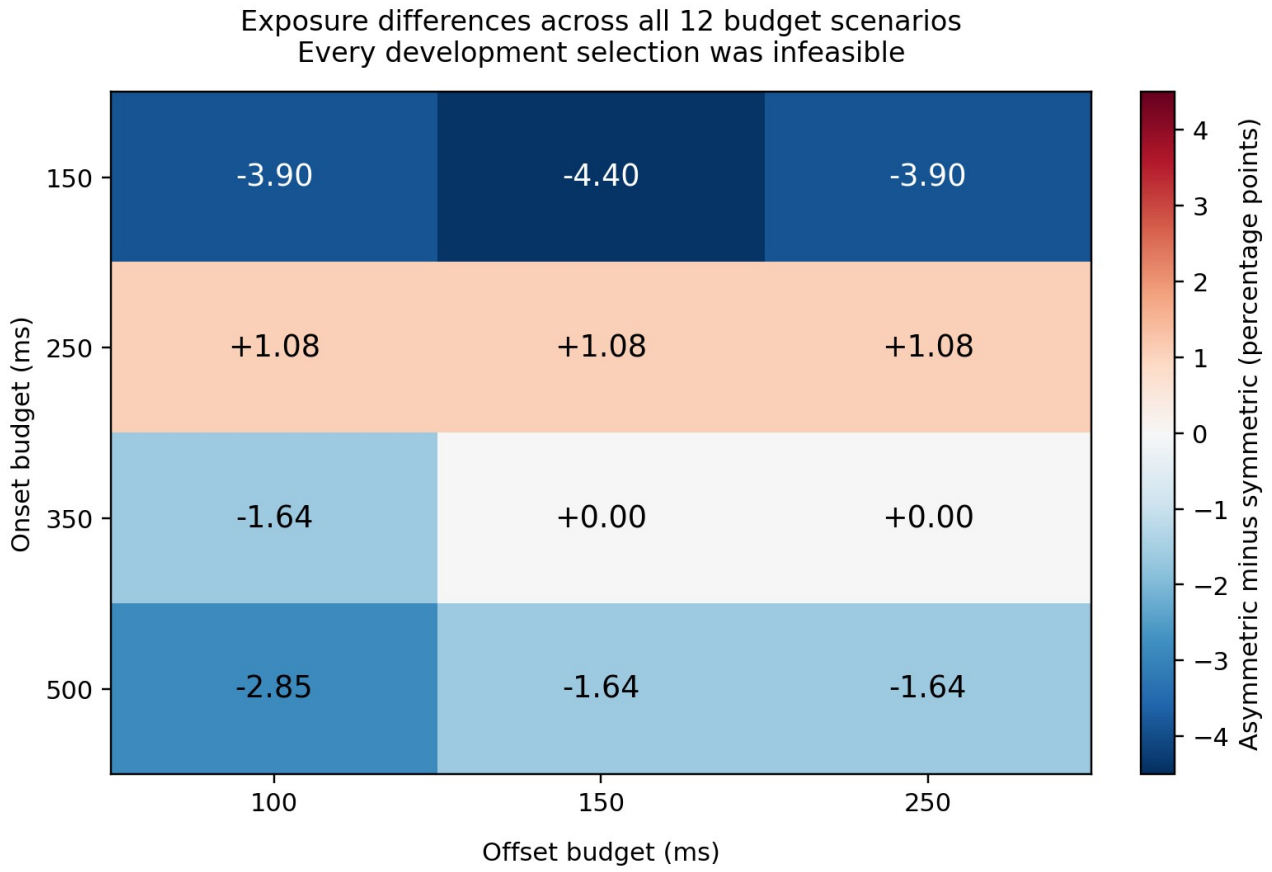
Figure 2 illustrates correct, wrong-active, and zero-command fractions, where each bar indicates equal-participant mean fractions of eligible DB5 active-interior time. Interiority is divided into sections for each type of activity: correct commands, wrong active commands, and inactive (zero output). The MAV-locking results illustrate why very high active coverage coincides with bad command accuracy.

Sensitivity analyses and exploratory adaptation

From 23 sensitivity analyses for DB5, seven methods generated a total of 1,288 participant-method outcome observations. None of the approaches managed a simultaneous success on any individual in any particular case. Different exposure values resulted from varying the selection budgets: the asymmetry condition reduced exposure by 4.398 points compared to symmetric at the onset and offset of 150 ms cells, but raised exposure by 1.077 points at the critical 250/150 ms cell. Note that these results correspond to different selections under different targets. The former did not make the latter possible. Figure 3 shows each budget cell.

In the 100 ms feature window, the difference between the asymmetric and symmetric conditions was +4.560 points. In both 200 ms and 250 ms windows, the difference was zero since the settings selected produced identical output responses. The fixed logistic regression produced +1.564 points, and all eight fits succeeded. For annotation shifts, the differences were all positive, between +0.989 and +1.151 points. Using the alternate stimulus reference value yielded a difference of +0.793 points. Modifying only the stability span had no impact on exposure, as it simply recomputed timings with constant commands. See the full sensitivity analysis in Supplementary Table S1.

The exploratory OTVL-inspired adaptation could successfully calibrate for all 10 DB5 individuals. On average across the 8 evaluation subjects, rest exposure was 0.062%, nonzero interior coverage was 17.85%, and correct interior output was 12.00%. While these numbers indicate infrequent activation in the adaptation, none should be interpreted as disconfirming clinical functionality of the published OTVL model.



Exposure difference by budget combination is shown in Figure 3 below. The twelve preset budget combinations considered are displayed. Each cell represents the difference between asymmetric and symmetric exposure in percentage points (rest exposure), evaluated for the corresponding development selections. Negative values represent lower exposure for the asymmetric settings, while all development selections were found infeasible, achieving none of the eight possible joint successes. No substitution exists here for the fundamental comparison under 250 ms onset and 150 ms offset budgets.

Native DB2 replication

All forty required DB2 source members cleared the compatibility check with no participant or eligible final event removed. None of the seven DB2 development families met all joint requirements. Reselection analysis found symmetric persistence for three decisions and asymmetric persistence for five activation decisions and seven release decisions.

With respect to the asymmetric versus symmetric exposure comparison using transported DB5 settings, the former totaled 37.301 percent versus 33.136 percent for the latter, producing a paired contrast of +4.165 percentage points (+3.463,+4.904). When reselection was performed, the respective means were 34.992 percent versus 32.420 percent, generating a contrast of +2.572 percentage points (+2.186,+2.963). Both contrasts were positive for all thirty-two final participants. Contrast results are summarized in Table 3, while timings and coverages are presented in Table 4.

Table 3 Paired exposure contrasts

Comparison	Participants	Difference pp	95% interval pp
DB5 primary	8	+1.077	0.764 to 1.397
DB2 transported	32	+4.165	3.463 to 4.904
DB2 reselected	32	+2.572	2.186 to 2.963

Difference in annotated-rest command exposure for asymmetric vs. symmetric strategies. Positive numbers indicate higher exposure. Paired-participant bootstrap percentiles are calculated with 10,000 replicates. All results are purely descriptive after the failure of feasibility studies. Both comparisons for DB2 use identical sets of participants. pp = percentage points.

Table 4 DB2 timing coverage and correctness

Method	Rest exposure %	Onset success %	Offset compliance %	Active coverage %	Correct interior %
Transported symmetric	33.14	18.11	28.40	87.29	58.01
Transported asymmetric	37.30	18.93	24.82	89.01	58.64
Reselected symmetric	32.42	16.73	29.04	87.27	59.53
Reselected asymmetric	34.99	15.17	25.55	88.95	61.50

Equal-participant means across 32 subjects. Each strategy resulted in 0/32 joint successes. Non-zero coverage does not necessarily imply proper performance of the commanded action. Transported and reselected configurations both arise from development families that failed the joint requirements.

The 19 comparisons using DB2 gave rise to zero out of 32 participants satisfying the joint criteria. The actual exposure proportions for DB2 were higher than those for DB5; nonetheless, this comparison could not partition the effects of acquisition, subject differences, annotation, or decoding performance. Replication applied temporally parametric rather than trained decoding models, whereas both DB2 comparisons made use of the same pool of participants.

Reproducibility and sampled computation

The full suite of unit checks passed successfully, satisfying 51 checks in total. Independent validation consisted of validating all 608 DB2 saved output streams and 41,344 stable event confirmations. The publicly released final comparison runner recreated the results of 112 DB5 and 608 DB2 participant method evaluations. The 84 DB5 and 114 DB2 reported summary metrics agreed with their archived values within numeric tolerances in the runner; participant effects and bootstrapped outputs of the three key contrasts agreed perfectly. All 138 publicly released tagged files matched the Git blobs they corresponded to.

Performance profiling sampled 11,200 DB5 and 22,400 DB2 updates. Maximum recorded complete update times were 20.312ms and 5.112ms, respectively, for neither of which did any sample exceed the 25ms hop timeout. These figures apply specifically to the observed hosts and sampled update paths. They are neither guaranteed worst cases nor end-to-end embedded controller latencies.

Discussion

There is no support for the predicted advantages of separate activation and release persistence when subject to the imposed requirements. Indeed, feasibility of development fails before feasibility of demonstrating an improvement over constraints in the first place. As expected, the descriptive fallback in DB5 extended command time during rest periods, while the two DB2 comparisons both increased command exposure in the same direction. This result highlights precisely what the design achieved under these conditions, successes and failures alike.

In particular, it bears noting how the asymmetrical design choice used immediate activation followed by a two-stage decision process for release. Contrast this approach against its fallback, in which a single decision sufficed for both purposes. Given the minimum possible 25ms hop, release persistence allows maintenance of the activation decision across short predicted rest spans. This is reflected in the observed increase in rest-command exposure. But such observation says little about whether asymmetric designs universally perform worse than symmetric designs. Budget cells selected alternative settings for different budgets. Moreover, many parameter sets swapped the exposure difference while failing to fulfill joint feasibility.

Any attempt to suppress rest must also incorporate an appropriate activity criterion. On the one hand, highly confident rejection decisions can substantially reduce rest exposure by rejecting much activity. On the other hand, MAV locking left many erroneous activity decisions locked. Both effects defy description according to any single favorable statistic. The fraction of correct commands, erroneous active commands, and zero-command decisions reveals this trade-off. Similarly, satisfaction of offset compliance must be carefully considered, as an already inactive fallback can meet the rest requirement without unlocking any real activity command.

Our conclusions agree with previous studies [1–7] rather than replacing them. Our contributions include the establishment of a common replay environment, per-subject feasibility checks for inclusion, native-time scoring of held commands, and tracking variations within preset parameter sets. We do not claim that temporal filtering is inherently insufficient for any application. Weak onset performance means failure cannot be assigned purely to the filter. The decoder model, annotations used, required confirmation duration, and selected deadlines jointly define this experiment.

DB2 extends analysis to cover variations in acquisition properties based on individual calibration. Concordance with the results above further corroborates this account of particular parameters. Nonetheless, results from two separate acquisition databases should not be pooled together as though they supplied 40 proven independent biological participants. Higher exposure in DB2 cannot be cited as definitive evidence that one control algorithm produces more desirable results. Several properties of the second acquisition method were altered simultaneously.

Future engineering analyses will require determination of feasible operation conditions before claiming improvements therein. This determination may demand a different decoder, a revised control task, or novel timing targets based on user feedback. Any such revision constitutes a new experiment. Results presented here cannot simply be overridden by retrospective selection of desired parameters. A user-in-the-loop test with actual actuator movement and pre-test calibration must then determine whether a beneficial trade-off indeed leads to improvement.

Limitations

The data came exclusively from intact limbs carrying out prescribed activities; amputee datasets, clinical patient datasets, closed-loop experiments, and real-world actuator tests were excluded. Specifically, our selected activity did not contain any active-to-active transitions within itself, so natural movement-to-movement switching could not be analyzed here. The publicly available labels were retroactively adjusted for refinement and do not actually represent intentions; instead, label changes served to probe sensitivity to temporal alignment but still would not uncover actual intentions themselves. Thus, it becomes clear why distinctions between refined and stimulus labels must persist since the differences arise from the unique properties of the dataset at hand.

Development sample sizes were relatively small overall, especially in the DB5 two-subject group. Requiring every member of the development set to satisfy three strict criteria could lead to unstable selections, while the criteria applied were engineering specifications and not evidence-based

standards regarding usability. The 100 ms confirmation window period uses up much of the allocated stopping allowance time budget of 150 ms. The failure described applies specifically to defined discrete grids and basic decoders, although the problem may still have solutions elsewhere.

Preventing the overlap between training and test raw ranges via raw-range separation solves the problem of leakage between owned ranges, but within-session correlation still remains. Training iterations after evaluation repetition two invalidates prospective analysis. Class priors were uniform, and the chosen featureset affects transitions and rest epochs. No confidence calibration nor comprehensive decoder exploration took place independently of those classifiers tested. Midpoint assignment of rest epochs, block resetting, omission of warmups, annotations' validity, and conditions of release all change the estimand. Correctness applies to the labeled movements alone, whereas coverage considers all active commands. Complete event diagnostics remain accessible in the analysis records so that distinction could be explored further if needed. Confidence bands are non-simultaneous bootstrap intervals and do not account for participant-level variability beyond what occurred within the available cohorts. The p-value descriptions offered cannot save feasibility from unmet criteria.

Lastly, biological independence between datasets was never proven here. An OTVL-style system architecture was implemented, albeit somewhat differently and in a highly exploratory fashion relative to its full version in prior work. Profiling finite hosts provides no guarantee regarding end-to-end responsiveness, tolerance to electrode malfunction, adaptation, or clinical viability—none of which were evaluated in the current project.

Conclusion

However, in this offline replay testbed, none of the controller grid combinations meet the aggregate criteria of onset, termination, and coverage. The main asymmetric default increased nonzero-command exposure during annotated rest, and both transported and independently selected DB2 configurations showed the same behavior. The conclusion here is obvious—these pipelines fail to satisfy the imposed constraints; see reproducible outputs defined below. Additional tests are necessary before assessing performance by actual users on real hardware in order to prove feasibility as a prosthesis controller.

Data and code availability

Our public codebase is hosted here: https://github.com/KSRICHARAN7/emg-command-filtering-benchmark, v0.11.0 release [12]. Data for the Ninapro studies come from the original repository [10,11]. Neither raw waveform data nor extracted features, fitted models, or sample-level command traces are included within the distribution of the publicly available code. Our public benchmarking code regenerates our core DB5 and DB2 comparisons but does not reproduce all experiments performed for the original development phase, including sensitivities or additional explorations. Details regarding these differences are provided within the release scope documentation.

Ethics and declarations

This analysis made use of publicly available, de-identified EMG recordings. Neither the recruitment of new subjects nor any interventions took place. Procedures regarding data collection and informed consent were detailed in prior publications [8,9]. No claims are made concerning institutional ethics board approval or exemption determination here.

AI assistance

GPT was only used for code debugging. The prose is completely retained by the author.

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Supplementary materials

The results presented here retain all specified sensitivities as well as both the main analysis and replication methodologies employed. These findings are purely descriptive, resulting from failure to meet joint feasibility requirements. Neither the methodology nor participants have undergone any renaming procedures. Duplicate results will not be considered as independent tests.

Supplementary Table S1 All DB5 sensitivity contrasts

Case	Difference pp	95% interval pp
annotation_shift_+100ms	+0.989	0.658 to 1.329
annotation_shift_+50ms	+1.039	0.724 to 1.362
annotation_shift_-100ms	+1.151	0.848 to 1.462
annotation_shift_-50ms	+1.119	0.814 to 1.430
annotation_stimulus	+0.793	0.429 to 1.201
budget_on150_off100	-3.901	-5.727 to -2.288
budget_on150_off150	-4.398	-6.456 to -2.562
budget_on150_off250	-3.901	-5.727 to -2.288
budget_on250_off100	+1.077	0.764 to 1.397
budget_on250_off150	+1.077	0.764 to 1.397
budget_on250_off250	+1.077	0.764 to 1.397
budget_on350_off100	-1.638	-2.279 to -1.059
budget_on350_off150	+0.000	0.000 to 0.000
budget_on350_off250	+0.000	0.000 to 0.000
budget_on500_off100	-2.850	-4.065 to -1.761
budget_on500_off150	-1.638	-2.279 to -1.059
budget_on500_off250	-1.638	-2.279 to -1.059
classifier_logistic	+1.564	1.186 to 1.925
stable_150ms	+1.077	0.764 to 1.397
stable_50ms	+1.077	0.764 to 1.397
window_100ms	+4.560	3.202 to 5.924
window_200ms	+0.000	0.000 to 0.000
window_250ms	+0.000	0.000 to 0.000

There were eight paired participants per condition. All seven procedures had 0/8 joint successes in every scenario considered. Note that bootstrap intervals are merely descriptive and cannot provide simultaneous confidence bounds. Also, the 250/150 ms budget allocation cell is a repetition of the main focus.

Supplementary Table S2 All DB5 primary methods

Method	Rest exposure %	Onset success %	Offset compliance %	Active coverage %	Correct interior %
Selected raw	7.22	6.25	37.87	93.30	57.76
Selected majority	7.22	6.25	37.87	93.30	57.76
Selected symmetric	7.22	6.25	37.87	93.30	57.76
Selected asymmetric	8.29	6.99	33.82	94.39	58.10
Selected confidence rejection	7.22	6.25	37.87	93.30	57.76
Selected DCIR	2.62	1.10	75.00	64.22	49.61
Selected MAV lock	1.90	1.10	66.91	78.70	18.22
Fixed raw	7.22	6.25	37.87	93.30	57.76
Fixed majority	7.62	1.10	33.09	94.01	61.09
Fixed confidence rejection	0.11	0.00	99.26	23.25	21.68
Fixed DCIR	0.34	0.37	94.85	41.62	37.79
Fixed MAV lock	2.07	0.74	63.24	84.68	24.31
Subset activation only	7.22	6.25	37.87	93.30	57.76
Subset release only	8.29	6.99	33.82	94.39	58.10

None of the fourteen streams produced any joint success (0/8). Those methods that were chosen and subsequently subset acted as infeasible placeholders, while the fixed methods provided predefined comparisons. Duplicate streams were kept under the same method name and thus provide no additional evidence.

Supplementary Table S3 All DB2 final methods

Method	Rest exposure %	Onset success %	Offset compliance %	Active coverage %	Correct interior %
Transported raw	33.14	18.11	28.40	87.29	58.01
Transported majority	33.14	18.11	28.40	87.29	58.01
Transported symmetric	33.14	18.11	28.40	87.29	58.01
Transported asymmetric	37.30	18.93	24.82	89.01	58.64
Transported confidence rejection	33.14	18.11	28.40	87.29	58.01
Transported DCIR	15.20	13.14	58.55	65.12	50.57
Transported MAV lock	1.79	5.88	80.88	54.29	23.11
Reselected raw	33.14	18.11	28.40	87.29	58.01
Reselected majority	31.75	12.32	29.41	87.22	61.01
Reselected symmetric	32.42	16.73	29.04	87.27	59.53
Reselected asymmetric	34.99	15.17	25.55	88.95	61.50
Reselected confidence rejection	12.84	12.32	62.32	63.07	48.55
Reselected DCIR	10.76	10.29	69.21	58.65	47.72
Reselected MAV lock	8.49	7.90	60.94	62.55	24.50
Fixed raw	33.14	18.11	28.40	87.29	58.01
Fixed majority	32.27	14.25	29.23	87.31	60.13
Fixed confidence rejection	0.95	1.84	95.68	23.78	21.98
Fixed DCIR	4.88	5.42	84.19	44.34	39.24
Fixed MAV lock	1.85	6.07	80.24	55.36	26.13

All 19 methods had 0/32 joint successes. The numbers above are equal-participant means calculated across 1,088 events. The transported/reselected methods act as infeasible proxies for development, whereas the fixed methods act as prespecified benchmarks.

Supplementary Table S4 Controller configuration grids

Family	Count	Configuration values
Raw	1	One unchanged raw decision stream.
Majority vote	6	Decision counts 1, 3, 5, 7, 9, 13.
Symmetric	6	One common count from 1, 2, 3, 5, 7, 9.
Asymmetric	36	Activation × release: all pairs from 1, 2, 3, 5, 7, 9; active changes use activation count.
Confidence rejection	10	Thresholds 0, .4, .5, .6, .7, .8, .9, .95, .97, .99.
DCIR	28	qmin .2/.4/.6 × qmax .8/.95/.989 × tau 100/250/500 ms; plus (.4,.989,480 ms).
MAV locking	20	Training-rest SD multipliers 1, 2, 3, 4 × initial votes 1, 3, 5, 6, 9.

107 settings were examined per development search. The identical finite grids were utilized during DB2 reselection. Full canonical parameter JSON, historical lock hashes, and the selected settings are provided in the publicly available release. These families constitute common-decoder implementations rather than reconstructions of entire published systems.